```
# Install OpenCV (only needed first time)
!pip install opencv-python

# Import libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
```

```
Requirement already satisfied: opencv-python in /usr/local/lib/python3.12/dist-packages (4.13.0.92)
Requirement already satisfied: numpy>=2 in /usr/local/lib/python3.12/dist-packages (from opencv-python) (2.0.2)
```

```
# Upload image from your computer
uploaded = files.upload()
```

Choose Files messi5 (1).jpg

messi5 (1).jpg(image/jpeg) - 12252 bytes, last modified: 2/24/2026 - 100% done

```
Saving messi5 (1).jpg to messi5 (1).jpg
```

```
# Replace with your uploaded file name
img = cv2.imread("/content/messi5 (1).jpg")

assert img is not None, "Image not found!"

# Convert BGR to RGB
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()
```

```
cropped = img[50:300, 100:400]

cropped_rgb = cv2.cvtColor(cropped, cv2.COLOR_BGR2RGB)

plt.imshow(cropped_rgb)
plt.title("Cropped Image")
plt.axis("off")
plt.show()
```

Cropped Image

```
resized = cv2.resize(img, (300, 300), interpolation=cv2.INTER_LINEAR)

resized_rgb = cv2.cvtColor(resized, cv2.COLOR_BGR2RGB)

plt.imshow(resized_rgb)
plt.title("Resized Image")
plt.axis("off")
plt.show()
```

Resized Image

```
rows, cols = img.shape[:2]

# Shift right 50, down 30
M_trans = np.float32([[1, 0, 50],
                      [0, 1, 30]])

translated = cv2.warpAffine(img, M_trans, (cols, rows))

translated_rgb = cv2.cvtColor(translated, cv2.COLOR_BGR2RGB)

plt.imshow(translated_rgb)
plt.title("Translated Image")
plt.axis("off")
plt.show()
```

Translated Image

```
rows, cols = img.shape[:2]
center = (cols // 2, rows // 2)

M_rot = cv2.getRotationMatrix2D(center, 45, 1)

rotated = cv2.warpAffine(img, M_rot, (cols, rows))

rotated_rgb = cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB)

plt.imshow(rotated_rgb)
plt.title("Rotated (45°)")
plt.axis("off")
plt.show()
```

Rotated (45°)

```
scaled = cv2.resize(img, None, fx=1.5, fy=1.5, interpolation=cv2.INTER_CUBIC)

scaled_rgb = cv2.cvtColor(scaled, cv2.COLOR_BGR2RGB)

plt.imshow(scaled_rgb)
plt.title("Scaled Image")
plt.axis("off")
plt.show()
```

Scaled Image

```
rows, cols = img.shape[:2]

M_shear = np.float32([[1, 0.5, 0],
                      [0, 1, 0]])

sheared = cv2.warpAffine(img, M_shear, (int(cols * 1.5), rows))

sheared_rgb = cv2.cvtColor(sheared, cv2.COLOR_BGR2RGB)

plt.imshow(sheared_rgb)
plt.title("Sheared Image")
plt.axis("off")
plt.show()
```

Sheared Image

```
flipped = cv2.flip(img, 1)

flipped_rgb = cv2.cvtColor(flipped, cv2.COLOR_BGR2RGB)

plt.imshow(flipped_rgb)
plt.title("Flipped Image")
plt.axis("off")
plt.show()
```

Flipped Image

```
rows, cols = img.shape[:2]

pts1 = np.float32([[50, 50],
                   [200, 50],
                   [50, 200]])

pts2 = np.float32([[10, 100],
                   [200, 50],
                   [100, 250]])

M_affine = cv2.getAffineTransform(pts1, pts2)

affine = cv2.warpAffine(img, M_affine, (cols, rows))

affine_rgb = cv2.cvtColor(affine, cv2.COLOR_BGR2RGB)

plt.imshow(affine_rgb)
plt.title("Affine Transformation")
plt.axis("off")
plt.show()
```

Affine Transformation

```
rows, cols = img.shape[:2]

pts1 = np.float32([[50, 50],
                   [cols-50, 50],
                   [50, rows-50],
                   [cols-50, rows-50]])

pts2 = np.float32([[0, 0],
                   [300, 0],
                   [0, 300]
```

```
                        [0, 300],
                        [300, 300]])

M_persp = cv2.getPerspectiveTransform(pts1, pts2)

perspective = cv2.warpPerspective(img, M_persp, (300, 300))

perspective_rgb = cv2.cvtColor(perspective, cv2.COLOR_BGR2RGB)

plt.imshow(perspective_rgb)
plt.title("Perspective Transformation")
plt.axis("off")
plt.show()
```

Perspective Transformation